

Abelian

$ab = ba \rightarrow \text{group} \rightarrow \Sigma$

$\frac{\langle a, b | ab=ba \rangle}{\mathbb{Z} \times \mathbb{Z}}$

$S^1 \vee S^2$

$T = \mathbb{Z} \times \mathbb{Z}$

$K(\cdot, \mathbb{Z} * \mathbb{Z})$

[illegible]

RP^2 : 0-cell: $\{v\}$
 1-cell: $\{a\}$
 2-cell: $\{a^2\}$

RP^2 : $\mathbb{Z}_2$ $\langle a \mid a^2 = 1 \rangle$

RP^2 : 0-cell: $\{v\}$
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$\pi_1(RP^2) = \mathbb{Z}_2$

Hand-drawn sketches on graph paper:

- A sequence of five circles arranged in a horizontal row.
- A flower-like shape with five petals inside a circle.
- A spiral shape with concentric loops.
- A horizontal line with four arrows pointing to the right.
- A complex plane with a branch cut along the real axis, indicated by a line with arrows and a '2'.
- A surface plot of a function, possibly a saddle shape, with arrows indicating direction.
- A contour integral in the complex plane, labeled with $\oint \frac{1}{z} dz$ and $\frac{2\pi i}{24}$.